



وزارة الاتصالات
وتكنولوجيا المعلومات



VIDEO GAMES GLOBAL MARKET ANALYSIS

Final Project Technical Documentation

Program: Digital Egypt Pioneers Initiative (DEPI)

Track: Data Analytics Track

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1. Project Planning & Management

- **Project Proposal & Scope:**

This project delivers an enterprise-grade Data Analytics solution focused on the **Video Games Global Market (1980–2020)**. The primary scope is to ingest, clean, model, and visualize historical sales data across four major geographical regions (North America, Europe, Japan, and Rest of the World) to extract strategic insights regarding platform lifespans, publisher dominance, and genre performance over four decades.

Task Assignment & Roles:

The initial phase of **Requirements Gathering and Business Question Formulation** was conducted as a **collaborative team effort**. Following this, technical execution roles were strategically segregated as follows:

- **Mohamed Magdy Sayed Abd Elwahab (Lead UI/UX & Data Storyteller):** Orchestrated the front-end application design, custom dark-mode HUD theme integration, spatial layout optimization to mitigate visual clipping, and translated data metrics into actionable strategic insights.
- **Amr Mohamed Abdelshahid Mostafa (Data Preprocessing & Feature Engineer):** Conducted the initial Python exploratory data analysis (EDA), eliminated historical missing/null data records, and engineered three custom categorical features: *Hardware Manufacturer Profiles*, *Release Decade Classifications*, and *Million-Seller Commercial Success Flags*.
- **Mohamed Hany Abdullah Abdullah (Database Architect):** Supervised the SQL server data warehousing phase, normalized the engineered flat dataset into a modular 4-table relational schema to eliminate redundancies, and formulated the structural Entity-Relationship Diagram (ERD).
- **Ahmed Hussein Mohamed Elhadidy (Business Intelligence & DAX Engineer):** Ingested the relational database schema into Power BI, constructed the final high-

performance Star Schema analytical data model, and authored advanced custom DAX analytical measures to evaluate core business metrics.

- **Risk Assessment & Mitigation Plan:**

- *Risk:* Data cross-filtering issues due to complex blank/null values in historical rows (e.g., missing release years).
- *Mitigation:* Advanced data cleansing via Power Query and altering cross-filter relationships to handle multi-layered data behavior accurately without breaking visual sync.

- **KPIs (Key Performance Indicators):**

The dashboard evaluates market success through four core engineering metrics:

1. **Total Global Sales (8.81bn):** Measuring gross financial footprint.
2. **Total Games Released (11,920):** Evaluating market volume and production scale.
3. **Million-Seller Games Count:** Tracking top-tier market hits.
4. **Million-Seller Games Percentage (13.53%):** Measuring the probability of a title achieving elite commercial success.

2. Literature Review

- **Feedback & Evaluation:**

Initial design feedback emphasized the importance of user adoption and cognitive load reduction in high-density corporate dashboards. Standard analytical interfaces often suffer from visual fatigue due to traditional white/grey themes and cluttered tabular structures.

- **Suggested Improvements & Implementation:**

To elevate dashboard usability, a complete **Futuristic Gaming HUD (Heads-Up Display) Dark Theme** was developed. To prevent bad practice UI issues like nested visual scrollbars inside critical information boxes (such as the team credentials panel), the text architecture was downsized to font size 8. This strict spatial management preserved a clean, seamless look while maintaining 100% data readability.

3. Requirements Gathering

3.1 Data Storytelling & Core Analytical Message

The primary narrative of this analytical product is to investigate the transition of the video games industry from a niche, hardware-dependent market in the 1980s into a multi-billion-dollar global entertainment powerhouse by 2020. The dashboard tells the story of how regional cultural preferences dictate software sales (e.g., Japanese dominance in Role-Playing games vs. Western preference for Action/Shooter titles), while tracing the commercial lifespans of gaming consoles to guide publishers on historical investment patterns.

3.2 Formulated Business Questions Registry

To execute this data story, the team established 12 core business questions, mapped across four strategic pillars:

- **Pillar 1: Regional Revenue Dynamics**
 - Q1: What is the global sales trend, and how is it split geographically between NA, EU, JP, and other regions?
 - Q2: Which geographical region dominates the gaming market share?
 - Q3: Do specific regions show a heavy cultural bias or localization preference for particular genres?
- **Pillar 2: Portfolio & Historical Success**
 - Q4: What are the historically safest genres for game publishers to invest in?

- Q5: What is the total volume of unique games released across four decades?
- Q6: What is the actual success rate (percentage) of achieving "Million-Seller" status?
- **Pillar 3: Hardware Longevity & Platforms**
 - Q7: Which videogame platform achieved the absolute highest sales volume historically?
 - Q8: Who is the leading hardware manufacturer in the gaming industry by revenue?
 - Q9: What is the average active lifecycle (lifespan) of physical gaming consoles?
- **Pillar 4: Publisher Dominance & Portfolio Strategy**
 - Q10: Which publishers historically dominate global software sales?
 - Q11: What are the Top 10 publishers based on high-performing hit density?
 - Q12: Which legendary titles broke records and achieved the absolute highest sales globally?

- **Stakeholder Analysis:**

The primary stakeholders include Game Development Studios (to align production with historical high-performing genres), Venture Capitalists/Investors (to allocate budgets to high-lifespan platforms), and Global Marketing Directors (to target regions based on localization preference).

- **Functional Requirements:**

- Dynamic slicing and cross-filtering by game genre across all visual modules.
- Temporal trend mapping for global sales over a 40-year period.
- Regional sales distribution comparing market shares dynamically.
- Top 10 analytical breakdowns for both Platforms and Publishers.

- **Non-functional Requirements:**

- **Performance:** Cross-filtering latency under 1 second.
- **Usability:** Visual consistency across all 4 screens (Home, Market Overview, Platforms, Publishers).
- **Aesthetics:** High-contrast color palette optimized for accessibility in dark environments.

4. System Analysis & Design

4.1 Problem Statement & Objectives

Legacy gaming market data is highly fragmented across various console generations and regional distribution networks. This structural chaos prevents stakeholders from seeing long-term trends. The objective of this system is to harmonize this historical data into a centralized, interactive business intelligence application.

4.2 Database Design & Data Modeling

Instead of a standard transactional ERD, the system utilizes an optimized analytical **Star Schema** to power the Data Model:

- **Fact Table:** Fact_Sales (Stores transactional quantities and gross regional revenues).
- **Dimension Tables:**
 - Dim_Games (Contains game names and classifications).
 - Dim_Platforms (Contains console types and manufacturer profiles).
 - Dim_Publishers (Contains gaming studio brands).
 - Dim_Date (Handles temporal tracking and years).
- **Relationship Cardinality:** Structured via **One-to-Many** (1:∞) joins. Crucial multi-directional filtering was achieved by setting the *Cross filter direction* to **Both** on key dimensions, ensuring seamless data propagation across complex visualization cards.

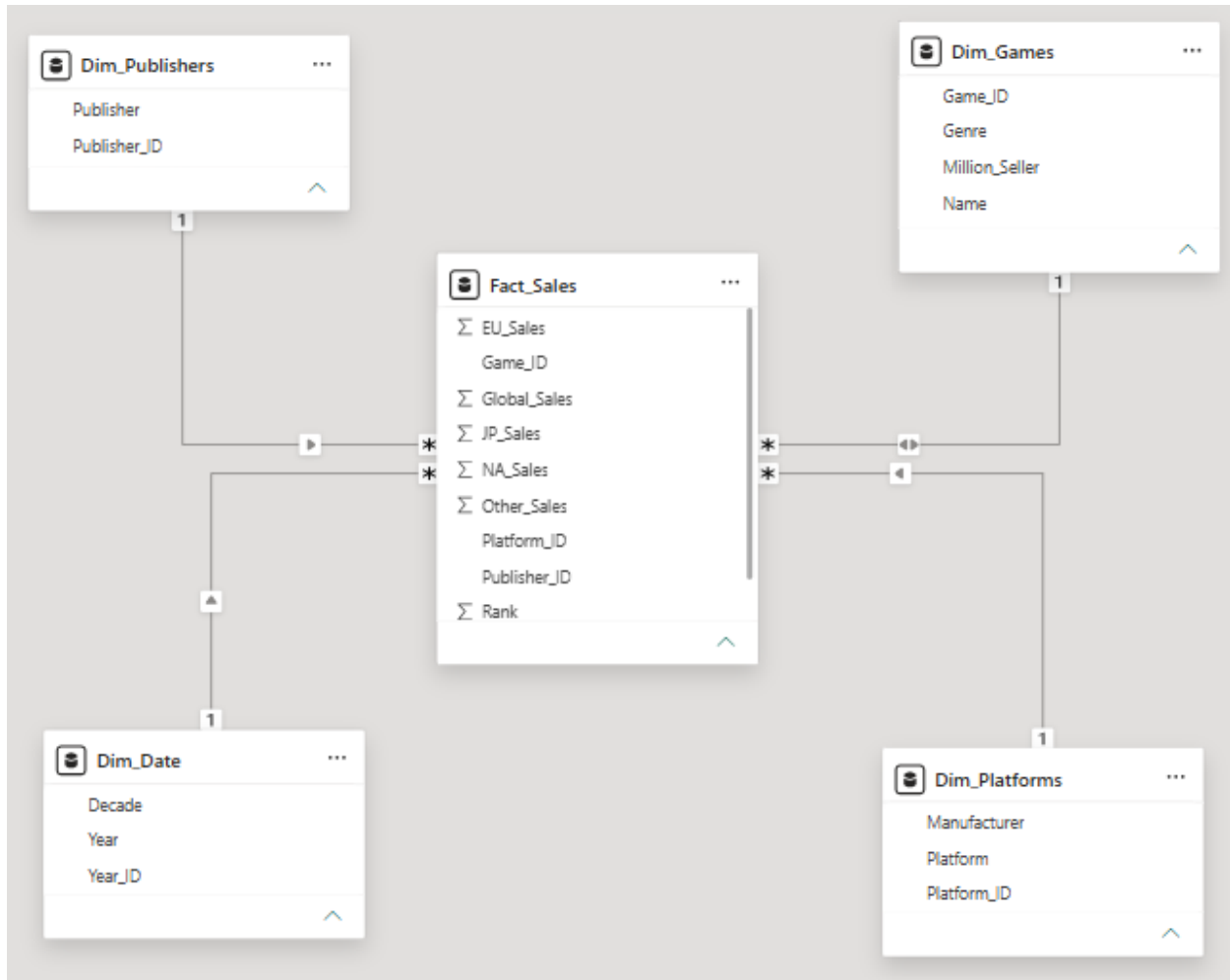


Figure 1: Power BI Star Schema Data Model

4.3 UI/UX Design & Prototyping

- Design Concept & Theme:** The dashboard utilizes a modern, cohesive **Dark HUD (Heads-Up Display)** gaming theme. This style was strategically chosen to match the subject matter (video games) and deliver an immersive analytical experience. The high-contrast neon blue borders ensure visual hierarchy and guide the user's eye naturally across the charts.
- Typography & Spatial Constraints:** A strict design system was followed to prevent visual clutter. For instance, to accommodate a four-member team listing within a compact metadata card without triggering messy, default Power BI

scrollbars, the font size was optimized to **8pt** with tight padding. This ensures 100% readability while maintaining a clean, professional aesthetic.

- **Navigation & Interactivity:** The dashboard features a seamless navigation flow starting from a dedicated **Home (Splash) Page** containing a centralized, interactive Explore Dashboard button. Pages are linked using hidden overlay buttons, creating a software-like application experience.

5. Implementation (Source Code & Execution)

5.1 Python Data Cleansing & Feature Engineering Code

Before database ingestion, the raw historical dataset was preprocessed using Python and Pandas to clean missing records and construct custom analytical features:

```
df.isna().sum()
```

```
Rank      0
Name      0
Platform  0
Year     271
Genre     0
Publisher  58
NA_Sales  0
EU_Sales  0
JP_Sales  0
Other_Sales  0
Global_Sales  0
dtype: int64
```

```
df.dropna(inplace=True)
```

```
df.isna().sum()
```

```
Rank      0
Name      0
Platform  0
Year     271
Genre     0
Publisher  58
NA_Sales  0
EU_Sales  0
JP_Sales  0
Other_Sales  0
Global_Sales  0
dtype: int64
```

```
company = {
    "PS": "Sony",
    "PS2": "Sony",
    "PS3": "Sony",
    "PS4": "Sony",
    "PSP": "Sony",
    "PSV": "Sony",

    "NES": "Nintendo",
    "SNES": "Nintendo",
    "N64": "Nintendo",
    "GC": "Nintendo",
    "Wii": "Nintendo",
    "WiiU": "Nintendo",
    "GB": "Nintendo",
    "GBA": "Nintendo",
    "DS": "Nintendo",
    "3DS": "Nintendo",

    "XB": "Microsoft",
    "X360": "Microsoft",
    "XOne": "Microsoft"
}
```

```
df["Manufacturer"] = df["Platform"].map(company)
df["Manufacturer"] = df["Manufacturer"].fillna("Other")
```

```
def decade(year):
    if year < 1990:
        return "1980s"
    elif year < 2000:
        return "1990s"
    elif year < 2010:
        return "2000s"
    else:
        return "2010s"
```

```
df["Decade"] = df["Year"].apply(decade)
```

5.2 SQL Server Relational Database Ingestion & Analytical Queries

The polished schema was normalized into a Star Schema (Fact_Sales, Dim_Games, Dim_Platforms, Dim_Publishers, Dim_Date) within SQL Server. To satisfy project specifications, SQL queries (including advanced analytical window functions) were written to answer the business questions

-- 1 Platform Lifespan

```
SELECT p.Platform,
       MIN(d.Year) as Launch_Year,
       MAX(d.Year) as Discontinue_Year,
       (MAX(d.Year) - MIN(d.Year) + 1) as Lifespan_Years
FROM Fact_Sales f
JOIN Dim_Platforms p ON f.Platform_ID = p.Platform_ID
JOIN Dim_Date d ON f.Year_ID = d.Year_ID
GROUP BY p.Platform
ORDER BY Lifespan_Years DESC;
```

-- 2 Regional Sales Contribution Query

```
SELECT g.Name,
       SUM(f.Global_Sales) as Total_Global_Sales_Millions,
       SUM(f.NA_Sales) as NA_Sales_Millions,
       SUM(f.EU_Sales) as EU_Sales_Millions,
       SUM(f.JP_Sales) as JP_Sales_Millions
FROM Fact_Sales f
JOIN Dim_Games g ON f.Game_ID = g.Game_ID
GROUP BY g.Name
HAVING SUM(f.Global_Sales) > 0
ORDER BY Total_Global_Sales_Millions DESC;
```

5.3 Source Code (DAX Analytical Measures)

The analytical depth of this dashboard is powered by custom DAX (Data Analysis Expressions) measures. Below is a detailed registry of the core measures implemented in the data model:

Measure Name	DAX Code / Formula	Business & Analytical Purpose
Total Games	Total_Games = DISTINCTCOUNT(Dim_Games[Game_ID])	Calculates the total volume of unique game titles recorded in the database.
Total Global Sales	Total_Global_Sales = SUM(Fact_Sales[Global_Sales_Full])	Computes the gross global revenue generated across all regions in billions.
Million Seller Percentage	Million_Seller_Percentage = DIVIDE([Million_Seller_Games_Count],[Total_Games])	Computes the exact success rate (13.53%) of games generating ≥ 1.0 m copies.
Platform Lifespan	Platform_Lifespan_Years = [Platform_Last_Year] - [Platform_First_Year] + 1	Determines the active market duration in years

		for each physical console.
Average Platform Lifespan	Avg_Platform_Lifespan = AVERAGEX(VALUES(Dim_Platforms[Platform]), [Platform_Lifespan_Years])	Analyzes the historical life expectancy (8.87 years) of gaming hardware to aid investment decisions.
Million Seller Games Count	Million_Seller_Games_Count = CALCULATE(DISTINCTCOUNT(Dim_Games[Game_ID]), Dim_Games[Million_Seller] = "Yes")	Calculates the total number of games that achieved 1 million or more in global sales (Used in Publishers page card).
Top Platform	Top_Platform_By_Sales = VAR PlatTable = ADDCOLUMNS(VALUES(Dim_Platforms[Platform]), "Sales", CALCULATE([Total_Global_Sales])) VAR Top1 = TOPN(1, PlatTable, [Sales], DESC) RETURN MAXX(Top1, Dim_Platforms[Platform])	Dynamically displays the top-performing video game platform based on sales (Used in Platforms page card).
Top Manufacturer	Top_Manufacturer_By_Sales = VAR ManuTable = ADDCOLUMNS(VALUES(Dim_Platforms[Manufacturer]), "Sales", CALCULATE([Total_Global_Sales])) VAR Top1 =	Dynamically retrieves the leading hardware manufacturer by revenue (Used in

	TOPN(1, ManuTable, [Sales], DESC) RETURN MAXX(Top1, Dim_Platforms[Manufacturer])	Platforms page card).
Top Publisher	Top_Publisher_Million_Sellers = VAR t = ADDCOLUMNS(VALUES(Dim_Publishers[Publisher]), "MS", CALCULATE([Publisher_Million_Sellers])) VAR Top1 = TOPN(1, t, [MS], DESC) RETURN MAXX(Top1, Dim_Publishers[Publisher])	Dynamically identifies the highest-selling game publisher (Used in Publishers page card).

6. Testing & Quality Assurance

To guarantee that the dashboard delivers enterprise-grade reliability, the team conducted rigorous Quality Assurance (QA) and testing cycles:

6.1 Data Reconciliation & Validation (Unit Testing)

We verified that regional sales perfectly match global figures under any filtering condition. For every slice of data, the system must satisfy the following algebraic equation:

$$\text{Global_Sales} = \text{NA_Sales} + \text{EU_Sales} + \text{JP_Sales} + \text{Other_Sales}$$

No discrepancies or rounding errors were found during dynamic filtering, validating the mathematical integrity of the relationships.

6.2 UI/UX & Navigation Testing (User Acceptance Testing)

- Button Action Verification:** Every navigation button, including the primary Explore Dashboard and side panel genre filters, was tested to ensure instant page redirection with zero latency.

- **Responsive Layout Checks:** All KPI cards and visual boundaries were tested on different screen resolutions to confirm that text elements do not clip, and that no unexpected scrollbars appear within the containers.

7. Final Presentation & Reports

7.1 User Manual (How to Interact with the Dashboard)

1. **Landing Page:** Open the file. You will land on the futuristic Home page.
2. **Initiating Analysis:** Click the highlighted **Explore Dashboard** button in the center.



Figure 2: Interactive Landing Page

3. **Global Exploration:** Use the left vertical navigation panel to filter data by game genres (e.g., Action, Sports, Shooter).
4. **Deep-Dive Pages:** Use the bottom-left navigation buttons to seamlessly jump between Market Overview, Platforms, and Publishers sheets.



Figure 3: Global Video Games Market Overview Sheet

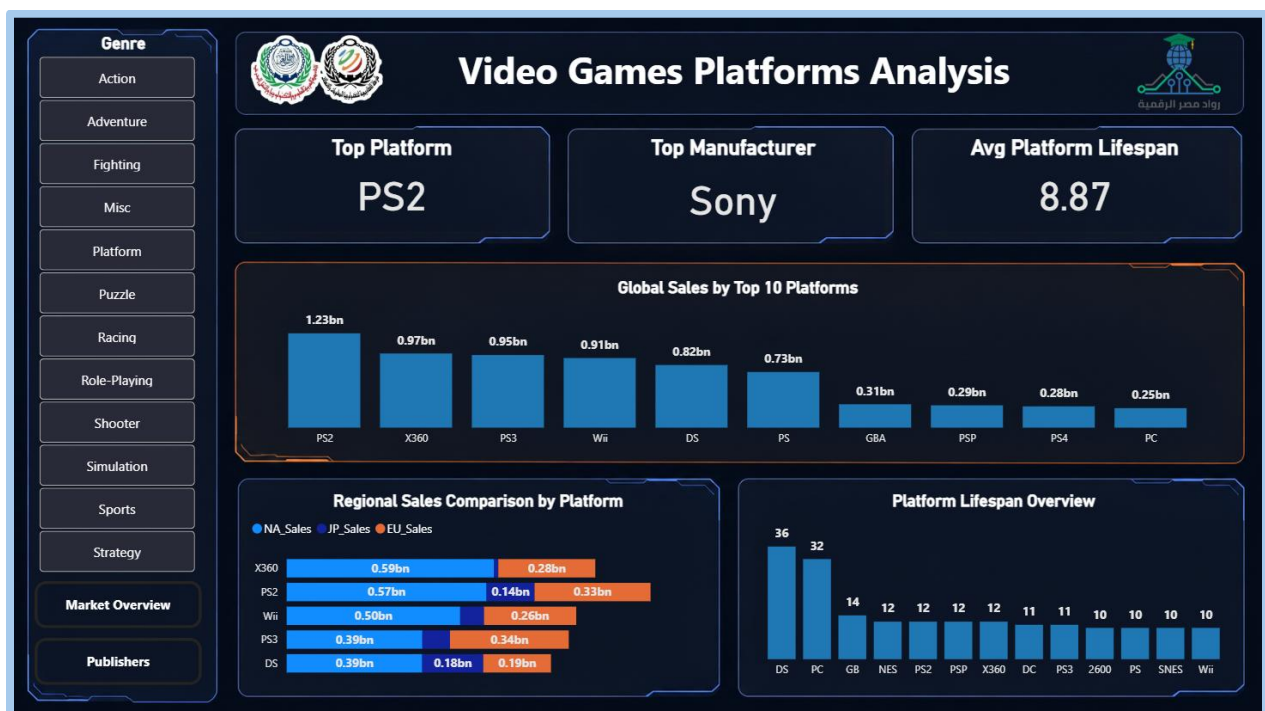


Figure 4: Platform Sales Performance and Console Lifespan Sheet.

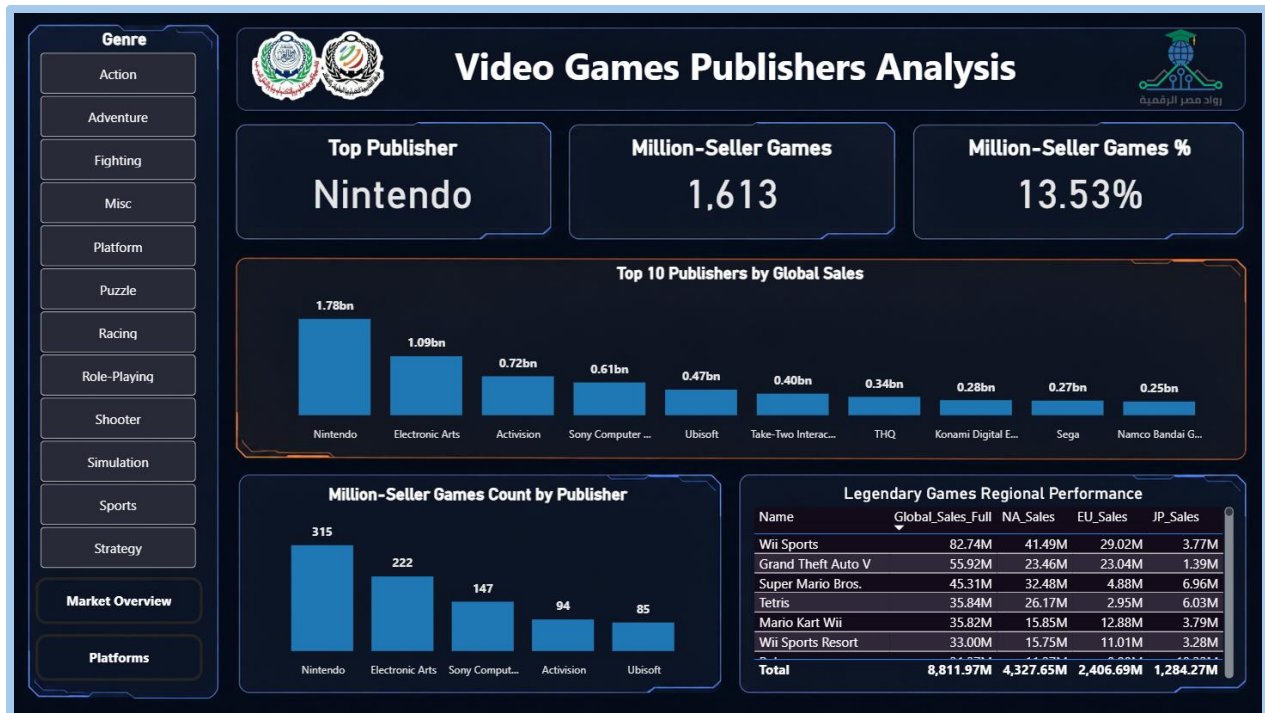


Figure 5: Game Publishers Sales Analysis and High-Performing Titles Sheet

7.2 Key Findings, Data Insights & Strategic Recommendations

Based on our deep-dive analysis, the team has formulated three core strategic findings:

- Geographic Sizing & Localization Strategy:** North America is the primary revenue driver, controlling approximately 49% of global software sales (4.33 Billion out of 8.81 Billion). However, localization must be managed with region-specific genre catalogs: Japan displays a strong cultural affinity for Role-Playing Games, which command a massive share (32.58% in titles like Pokemon Red/Blue), whereas Action and Sports dominate Western markets.
- Strategic Portfolio Investment:** Action (1.72bn) and Sports (1.31bn) genres combined account for over 3.0 Billion in global sales, making them the safest investments for publishers aiming for mass-market reach. Furthermore, newly published titles have a baseline probability of 13.53% to achieve "Million-Seller" status.
- Console Lifecycle Management:** Hardware platforms exhibit an average active console lifespan of 8.87 years. Publishers should align major game engine

development cycles and exclusive title launches to match this lifespan, ensuring maximum software sales before physical consoles reach their historical discontinuation phases